1. Allow a user to enter a quantity of an item. If the quantity is greater than or equal to 1000, the unit price should be $3.00. For quantities under 1000 the unit price is $5.00. Compute extended price to be quantity x unit price. Compute tax to be 7% of the extended price. The total is computed as an extended price plus the tax. Display the quantity, unit price, extended price, tax and total.

INPUT	Process	OUTPUT
Quantity as int Unitprice as float Extendedprice as float Tax as float Total as float	If quantity >= 1000, then unit price = 3, else 5 Extended price = unitprice*Quantity Tax = .07 * extended price Total = tax + extended price	‘Quantity:’ quantity ‘Extended price:’ extendedprice ‘Unit price: ‘ unitpriceLas ‘Tax:’ tax ‘Total:’ total

2. The program asks the user for an item and quantity. Determine the unit price of the item based on the chart below. Compute the extended price to be quantity x unit price. Display the item, unit price and extended price. Note: if the item entered is not A then assume the item is B. No need to check for B.
Item Unit Price
A $10.00
B $20.00
(Note: assume the user will enter the data correctly. Assume if they enter capital A then $10.00 gets assigned to unit price variable. Any other entry is assumed to be a capital B whether they enter B or not. Therefore, you only need a relational condition for A.)

```
if item == “A”:
Unit_price = 10.00
else:
Unit_price = 20.00
```

INPUT	Process	Output
String item (a or b) Int quantity Float price Float total	If item == ‘a’, then price = 10, else price = 20. Total = price*quantity	‘Total =’ total

3. Enter the number of books to order and cost per book. If the order total is over $50.00 shipping is free. If the order total is $50.00 or under charge $25 shipping. Display the order total and shipping charge (note 0 should display for a free shipping charge).

Input	process	output
Float cost Int books Float total Float shipping	Total = cost*books If total > 50, shipping = 0, else shipping = 25.	‘Total:’ total ‘Shipping: ‘ shipping

4. The warranty of an appliance depends on the cost of the appliance. For appliances over $1,000 the warrantee cost is 10% of the price. For appliances $1,000 or less the warranty cost is 5% of the price. The user will enter the name and cost of an appliance. Display name and cost of appliance, the cost of the warrantee and the total (cost of the appliance + warranty).

input	process	Output
String name Float cost Float warranty	If cost >1000, then warranty = .1*cost, else warranty = .05*cost Total = warranty+cost	‘Name ‘ name ‘Cost ‘ cost ‘Warrantee cost ‘ warranty ‘Total ‘ total

5. Enter the user’s last name, number of dependents and gross income. Compute adjusted gross income to be gross income minus dependents times $12000. Next determine an income tax rate. Adjusted gross incomes over $50,000 have a tax rate of 20%. Adjusted gross incomes $50,000 or under have a tax rate of 10%. Once you determine the tax rate, compute income tax to be adjusted gross income times tax rate. If the income tax is less than 0, set the income tax to $100. Display last name, gross income, number of dependents, adjusted gross income, and income tax

input	process	output
String name Int deps Float gross Float adjgross Float inctax	Adjgross = gross-(deps*12000) If adjgross > 50000, then inctax = .2*adjgross, else inctax=.1*adjgross. If adjgross < 0, then Inctax = 100	‘Last name ‘ name ‘Gross income ‘ gross ‘Number of dependants ‘ deps ‘Adjusted gross income ‘ adjgross ‘Income tax ‘ inctax